

SUMMARY

A Python-focused Software Engineer with 4+ years of experience delivering secure, cloud-native platforms across insurance and healthcare, specializing in microservices, data-intensive APIs, and real-time systems using Python (FastAPI, Flask, Django), React, and AWS (EKS, Lambda, RDS). Expert in architecting multi-tenant policy/claims systems, decomposing monoliths, and enforcing OAuth2/JWT/RBAC with ISO/HIPAA controls to improve release cadence, resilience, and data privacy. Proven impact includes 50% faster feature lead time, 35% fewer production incidents, 45% faster data lookups, 99.9%+ uptime, and 40% MTTR reduction through observability (Prometheus, Grafana, ELK) and SRE practices. Strong CI/CD ownership with GitHub Actions/Jenkins, blue/green and canary deployments, Terraform IaC, and container orchestration with Docker/Kubernetes on AWS. Hands-on with event-driven patterns (Kafka, Redis Streams, Celery, WebSockets) and data stores (PostgreSQL, MongoDB, Redis, S3/Parquet) for high-volume claims, billing, and connected-device workflows. Collaborates closely with cross-functional teams and stakeholders to translate business requirements into reliable, compliant software, consistently driving measurable, revenue-impacting outcomes.

SKILLS

Programming and paradigms:	Python (advanced), TypeScript/JavaScript; OOP, Functional Programming.
Backend and API frameworks:	FastAPI, Flask, Django; RESTful APIs, OAuth2, JWT, RBAC; FHIR/HL7 adapters, SMART-on-FHIR.
Frontend and data visualization:	React.js, Angular, D3.js.
Eventing and real-time systems:	Kafka, Redis Streams, WebSockets, Celery.
Datastores and data formats:	PostgreSQL, MongoDB, Redis; S3; Parquet.
Microservices and architecture:	Service decomposition, asynchronous services, domain-driven design; idempotency, circuit breakers; saga/outbox.
Cloud, containers, and IaC:	AWS (EKS, Lambda, RDS, S3, IAM, CloudWatch); Docker, Kubernetes (EKS), Helm; Terraform.
CI/CD and version control:	Git, GitHub Actions, Jenkins; blue/green and canary deployments.
Testing, quality, and observability:	PyTest, Behave (BDD), Selenium/Selenium Grid, pact (contract testing), Locust; Prometheus, Grafana, ELK; OpenTelemetry; Pydantic (v1/v2) for validation.
Security, compliance, and domain knowledge:	ISO/IEC 27001, HIPAA; encryption in transit/at rest; Secrets Manager/Vault; SAST/DAST; Stripe, Twilio; claims lifecycle, policy/billing, connected devices/mobile protection, renters/global housing.

WORK EXPERIENCE

Assurant, AZ | Software Engineer

July 2025 – Current

- Architected a multi-tenant policy/claims core with Python (Flask, Tornado) and React micro-frontends to standardize product setup across mobile protection and renters lines, accelerating B2B2C launches by cutting feature lead time 50%.
- Orchestrated secure gateways with Django REST Framework, OAuth2/JWT, and granular RBAC to enforce ISO/IEC 27001-style controls for partner/device trade-in portals, reducing unauthorized access incidents by 80%.
- Decomposed monolith billing and warranty flows into FastAPI microservices with Node.js event workers to uplift release cadence, improving deployment velocity and lowering prod incidents by 35%.
- Activated real-time notifications for claim status and device diagnostics via WebSockets, Redis Streams, and Celery to shrink support SLAs, lifting customer engagement by 20% in device and auto journeys.
- Optimized data access by modeling PostgreSQL OLTP and MongoDB for device logs with Redis caching, accelerating quotes/claims/account lookups by 45% to speed underwriting and servicing.
- Provisioned resilient AWS stacks (EKS, RDS, ElastiCache) using Docker and Terraform with blue/green rollouts and autoscaling, sustaining 99.9%+ uptime for consumer portals and partner APIs.
- Streamlined delivery via GitHub Actions CI/CD integrating unit, contract, SAST/DAST gates to compress build-and-release from 60 min to under 10 min, improving change failure rate.
- Instrumented observability with Prometheus, Grafana, and ELK to detect anomalies early and guide remediation, reducing MTTR by 40% across high-volume protection programs.
- Curated analytics backends in Python (Pandas/FastAPI) and React/D3 dashboards for Global Lifestyle/Housing KPIs, translating telemetry into actions that drove revenue-impacting optimizations.
- Governed compliance and data privacy with audit trails, encryption-at-rest/in-transit, and policy-as-code checks, safeguarding PII while enabling rapid product iteration across markets.

Assurant, AZ | Software Engineer Intern

March 2025 – June 2025

- Prototyped a lightweight claim intake microservice in Python (FastAPI) with Pydantic schemas and JWT auth to validate renter policy submissions, accelerating partner sandbox onboarding and de-risking APEX API integrations.
- Implemented event-driven data loaders using Python, Redis, and Celery to normalize mobile repair/work order feeds, improving downstream claim triage accuracy and shortening support touch-times in Connected Living.
- Containerized a Node.js notification worker and Python rules engine with Docker and Helm for EKS dev clusters, enabling reproducible environments and cutting defect reproduction time in Global Lifestyle sprints.
- Authored Grafana dashboards and Prometheus alerts for policy and claim latencies, revealing bottlenecks that reduced MTTR in pre-prod by driving targeted fixes across partner API flows.

- Developed React + D3 internal insights pages backed by a Python (Pandas/FastAPI) service to visualize quote-to-bind funnels for renters products, informing iteration priorities for revenue-impacting journeys.

Mphasis, India | Software Engineer

May 2020 – Aug 2023

- Engineered healthcare claim lifecycle modules using Python (Django, Flask) and React.js to embed audit trails and RBAC, ensuring HIPAA compliance; this was critical to accelerate provider onboarding efficiency by 30% and optimize reimbursement cycles by 45%, fulfilling Mphasis’ domain-driven focus on service transformation in healthcare.
- Integrated scalable FHIR/HL7-compliant APIs with FastAPI and automated ICD-10/CPT/NPI validations via Pydantic, to align with payer standards; this was executed to strengthen regulatory adherence and increase clinical data accuracy by 60%, advancing Mphasis’ mission of customer-centric transformation.
- Deployed asynchronous backend services with Python + Node.js for patient reminders and EDI file handling to minimize latency; this decision reduced backend processing time by 33% and lowered claim denial rates by 25%, thereby amplifying operational efficiency in healthcare ecosystems where speed-to-care is critical.
- Architected cloud-native microservices using Docker, Kubernetes (AWS EKS) for telehealth scheduling solutions; this design enabled seamless scalability and 99.95% uptime, ensuring clinical systems stayed resilient, in line with Mphasis’ Front2Back™ approach of scaling transformation through cloud innovation.
- Automated compliance-enforced deployments by building CI/CD pipelines with GitHub Actions & AWS CodePipeline; this eliminated manual release bottlenecks and cut deployment cycles from 120 minutes to just 15 minutes, aligning with Mphasis’ goal of driving speed-to-market transformation across business-critical platforms.
- Structured unstructured healthcare data by designing schema-flexible repositories in MongoDB Atlas, specialized for SOAP notes and diagnostic records; this effort improved provider query response times by 35%, empowering clinicians with faster access to actionable patient insights, directly enabling AI-powered decision-making.
- Validated software quality by automating regression and compliance test suites with Behave (BDD) and Selenium Grid, embedding them into cloud-based CI/CD; this approach reduced manual QA efforts by 40% while improving bug detection, which translated to stronger patient-data safety compliance in regulated healthcare domains.
- Presented healthcare KPIs, HEDIS metrics, and AI-powered predictions through intuitive dashboards using React.js + D3.js; by bridging data storytelling with technical logic, stakeholder change requests declined by 20%, reinforcing Mphasis’ innovation charter to be the “Driver in Driverless Car” for data-informed healthcare enterprises.
- Consolidated payer-provider interoperability by standing up SMART-on-FHIR endpoints with FastAPI and HL7 adapters, enabling secure EMR app embedding to expand data liquidity; this was done to streamline clinical exchanges and value-based workflows, elevating decision quality and interoperability compliance across payer operations.
- Fortified platform reliability and scalability by codifying cloud patterns with Terraform modules and EKS add-ons aligned to Mphasis Front2Back cloud guidelines, enabling right-sized autoscaling and fault isolation; this was pursued to standardize cloud-native execution and accelerate AI-enabled service transformation, improving resiliency for claims and telehealth workloads.

EDUCATION

Masters in Information Technology Northern Arizona University, AZ	2023 - 2025
Bachelors of Engineering in Computer Science Aarupadai Veedu Institute of Technology (AVIT), India	2017 - 2022